# Can Bureaucrats Solve NP-Complete Problems (and Do They Even Want To)?

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# Motivation (SNAP)

**Supplemental Nutritional Assistance Program**

- SNAP covers $\sim$ 12% of the population (about $\sim$ 0.4% of GDP)
- 1996: Introduce "work requirements" for "ABAWDs"
- "Areas" with "insufficient jobs" can apply to USDA-FNS for waivers

**Why this is a good laboratory**

- What constitutes an "area" or "insufficient jobs" is ambiguous
- 50+ decision makers over $\sim$30 years

# A complex regulatory environment

**Two statutory prongs, several operational routes**

- **Statutory threshold**
  - **10% Rule:** area unemployment $>$ 10% (12-mo, 3-mo) self-certifiable (survives the OBBB)
- **“Insufficient jobs”**
  - **LSA:** If DoL assigns this BLS unit (county) to be a “Labor Surplus Area”
  - **EB trigger:** Applies statewide if state is on the BLS “trigger list” for extended unemployment benefits
  - **20% Rule:** If the avg. unemployment rate over 24 months in an “area” is 20% above the national average in the same 24-month window.
  - Some federal exemptions (ARRA, FFCRA)

# The 20% rule and room for computational sophistication

**State agency's choices**

1. Choice of 24-month data window (starting after Jan of 2/3 FYs prior)
2. Choice of application date and waiver start date
3. Choice of (contiguous) LAUS units
4. Choice of group construction
5. Choice of BLS data vintage
6. Rounding rules ...

# The math of the weighted set-packing problem

**Primitives (fix a state $s$ and a 24-month window)**

- Counties $i \in I$: labor force $L_i$, unemployed $U_i$, policymaker value $V_i$.
- Threshold $\alpha = 1.2\,\tilde{\alpha}$ ($\tilde{\alpha}$ = national rate); slack $s_i := U_i - \alpha L_i$.

**The state's problem**

$$\max_{G \in \mathcal{P}(\mathcal{P}(I))} \sum_{g \in G} \sum_{i \in g} V_i \quad \text{s.t.} \quad g \text{ contiguous}, \;\; g \cap g' = \varnothing, \;\; \textstyle\sum_{i \in g} s_i \geq 0 \qquad \forall\, g \neq g' \in G$$

# Notions of optimality – Preferences?

**Valuation schemes $V^j$ (increasing sophistication)**

- $R^{\mathrm{N}}$: $V_i = 1$ $\quad R^{\mathrm{pop}}$: $V_i =$ population $\quad R^{\mathrm{SNAP}}$: $V_i =$ SNAP recipients

**Valuation ratio**

$$R^j_{st} := \frac{V^{j,\mathrm{obs}}_{st}}{V^{j,*}_{st}} = \frac{\sum_{g \in G^{\mathrm{obs}}} \sum_{i \in g} V^j_i}{\sum_{g \in G^*(V^j)} \sum_{i \in g} V^j_i} \in [0, 1]$$

- What is the difference between preferences and sophistication?
- Inferring preferences from revealed choice data?
- Pareto weights that rationalize observed choice?
- Are excluded counties correlated on observables?
- Counterfactual sophistication?